



PRIME

suspects

A *prime* number is a whole number that you can't divide into other whole numbers except for 1.

The number 23 is prime, for instance, because nothing will divide into it without leaving a remainder. But 22 isn't: 11 and 2 will divide into it. Some mathematicians call prime numbers the building blocks of maths because you can create *all other* whole numbers by multiplying primes together. Here are some examples:

$$55 = 5 \times 11$$

$$75 = 3 \times 5 \times 5$$

$$39 = 3 \times 13$$

$$221 = 13 \times 17$$

31 is prime

331 is prime

3331 is prime

33331 is prime

333331 is prime

3333331 is prime

33333331 is prime

But what about 3333333331

It turns out not to be, because:

$$17 \times 19607843 = 333333331$$

Which just goes to show that you can never trust a pattern just because it *looks like* it might continue. Mathematicians always need **proof**.

An unsolved mystery

The *mysterious* thing about primes is the way they seem to crop up at **random** among other numbers, without any pattern. Mathematicians have struggled for years to find a pattern, but with no luck. The lack of a pattern means prime numbers have to be hunted down, one by one.

HUNTING FOR PRIMES

	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Small primes are easy to hunt by using a "sieve". To do this, write out the numbers up to 100 in a grid, leaving out the number 1 (which isn't prime). Cross out multiples of two, except for 2 itself. Then cross out multiples of 3, except for 3. You'll already have crossed out multiples of 4, so now cross out multiples of 5, then multiples of 7. All the numbers left in the grid (coloured yellow above) will be prime.

2

3

5

7

11

13

17

19

23